Advanced Optics Laboratory (PH4201)

Experiment: 02

Determination of Polarization properties of Spatial Light Modulator (SLM) using Stokes Vector Polarimetry

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1 | Introduction

Optical polarimetry, ever since its discovery, has played important roles in gaining fundamental understanding of the nature of electromagnetic waves and answering some of the key questions related to the physics of light. Traditional polarimetry has also long being pursued for numerous practical applications in various branches of science and technology. Quantification of protein properties in solutions, testing of chiral (asymmetric) purity of pharmaceutical drugs, remote sensing in meteorology and astronomy, optical stress analysis of structures, and crystallography of biochemical complexes are just a smattering of its diversified uses. This particular polarimetry experiment involves determination of two important polarization characteristics of medium, namely, the linear retardance and optical rotation (defined subsequently), which has many practical applications.

2 | Objective

The main ojective of this experiment is to determine of linear retardance and Optical rotation of a Spatial Light Modulator (SLM) measuring various Stokes vector elements

3 | Equipment Required

The following equipments are required for this experiment

- Diode laser (650 nm)
- Spatial Light Modulator (SLM)
- Two Polarizer with mounts
- Photo-detector
- Optical table and mounts
- Computer with software for data acquisition and analysis
- Power supply for the SLM
- Glass plate with stage having rotational circular scale(in degree)

4 | Theory

4.1 | Basics of Polarization

Christian Huygens was the first to suggest that light was not a scalar quantity based on his work on the propagation of light through crystals; it appeared that light had 'sides' in the words of Newton. This vectorial nature of light is called polarization [2].

In a Cartesian system the components $u_x(\mathbf{r}, t)$ and $u_y(\mathbf{r}, t)$ are said to be the transverse components, and the component $u_z(\mathbf{r}, t)$ is said to be the longitudinal component when the propagation is in the z direction. Thus we can write

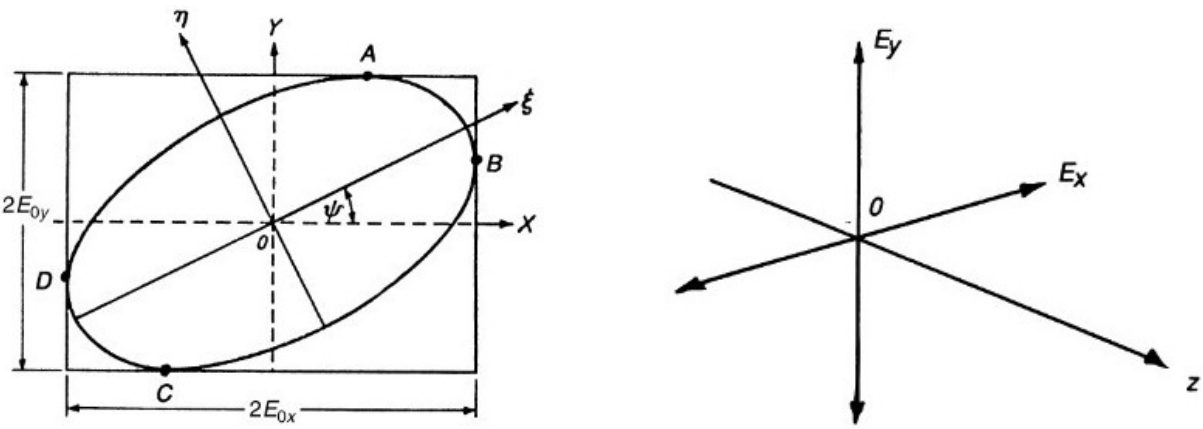
$$u_x(\mathbf{r}, t) = u_{0x} \cos(\omega t - \mathbf{k} \cdot \mathbf{r} + \delta_x), \tag{4.1a}$$
$$u_y(\mathbf{r}, t) = u_{0y} \cos(\omega t - \mathbf{k} \cdot \mathbf{r} + \delta_y), \tag{4.1b}$$
$$u_z(\mathbf{r}, t) = u_{0z} \cos(\omega t - \mathbf{k} \cdot \mathbf{r} + \delta_z). \tag{4.1c}$$

Experimentally found that light consisted only of the transverse components Equation (4.1a) and Equation (4.1b). If we take the direction of propagation to be in the z direction, then the optical field in free space must be described only by Equation (4.1a) and Equation (4.1b). where u_{0x} and u_{0y} are the maximum amplitudes and δ_x and δ_y are arbitrary phases. These two Equation (4.1a) and Equation (4.1b) are are spoken of as the polarized or polarization components of the optical field. Now, the transverse components are represented by

$$E_x(z,t) = E_{0x}\cos(\tau + \delta_x), \tag{4.2a}$$
$$E_y(z,t) = E_{0y}\cos(\tau + \delta_y), \tag{4.2b}$$

where $\tau = \omega t - kz$ is the propagator. The subscripts x and y refer to the components in the x and y directions. E_{0x} and E_{0y} are the maximum amplitudes, and δ_x and δ_y are the phases, respectively. As the field propagates, $E_x(z,t)$ and $E_y(z,t)$ give rise to a resultant vector. This vector describes a locus of points in space, and the curve generated by those points will now be derived. In order to do this Equation (4.2a) and Equation (4.2b) are written as

$$\frac{E_x}{E_{0x}} = \cos\tau\cos\delta_x - \sin\tau\sin\delta_x, \tag{4.3a}$$
$$\frac{E_y}{E_{0y}} = \cos\tau\cos\delta_y - \sin\tau\sin\delta_y. \tag{4.3b}$$



(a) An elliptically polarized wave and the polarization ellipse

(b) Propagation of the transverse optical field

Figure 4.1: Polarization of light

Hence,

$$\frac{E_x}{E_{0x}}\sin\delta_y - \frac{E_y}{E_{0y}}\sin\delta_x = \cos\tau\,\sin(\delta_y - \delta_x), \tag{4.4a}$$
$$\frac{E_x}{E_{0x}}\cos\delta_y - \frac{E_y}{E_{0y}}\cos\delta_x = \sin\tau\,\sin(\delta_y - \delta_x). \tag{4.4b}$$

Squaring (4.4a) and (4.4b) and adding gives

$$\frac{E_x^2}{E_{0x}^2} + \frac{E_y^2}{E_{0y}^2} - 2\frac{E_x}{E_{0x}}\frac{E_y}{E_{0y}}\cos\delta = \sin^2\delta, \tag{4.5a}$$
$$\delta = \delta_y - \delta_x. \tag{4.5b}$$

Equation (4.5a) is recognized as the equation of an ellipse and shows that at any instant of time the locus of points described by the optical field as it propagates is an ellipse (see Figure 4.1a). This behavior is spoken of as optical polarization, and Equation (4.5a) is called the *polarization ellipse* [2].

Also, we can write some more parameters of ellipse in terms of δ and α where $\tan\alpha = E_{0y}/E_{0x}$

$$\tan 2\psi = (\tan 2\alpha)\cos\delta, \qquad 0 \le \psi \le \pi, \tag{4.6a}$$
$$\sin 2\chi = (\sin 2\alpha)\sin\delta, \qquad -\frac{\pi}{4} \le \chi \le \frac{\pi}{4}. \tag{4.6b}$$

4.2 | Stokes Vector and Muller Matrix

The *Stokes vector* is a convenient method for describing the polarization state of light using measurable intensity quantities. It consists of four parameters (S_0, S_1, S_2, S_3), where S_0 represents the total intensity, while S_1, S_2, and S_3 characterize the linear and circular polarization components. Since these parameters can be obtained experimentally using polarizers and wave plates, the Stokes formalism is widely used in polarization analysis and optical characterization [1].

We now write the quantities inside the parentheses as

$$S_0 = E_{0x}^2 + E_{0y}^2, \tag{4.7a}$$
$$S_1 = E_{0x}^2 - E_{0y}^2, \tag{4.7b}$$
$$S_2 = 2E_{0x}E_{0y}\cos\delta, \tag{4.7c}$$
$$S_3 = 2E_{0x}E_{0y}\sin\delta, \tag{4.7d}$$

and then using the Equation (4.5) we can write

$$S_0^2 = S_1^2 + S_2^2 + S_3^2. \tag{4.8}$$

Also the other parameters of the polarization ellipse can be write interms of Stokes vectors

$$\tan 2\Psi = \frac{S_2}{S_1}, \tag{4.9a}$$
$$\sin 2\chi = \frac{S_3}{S_0}. \tag{4.9b}$$

And the degree of polarization are defined as following

$$P = \frac{I_{\text{pol}}}{I_{\text{tot}}} = \frac{\left(S_1^2 + S_2^2 + S_3^2\right)^{1/2}}{S_0}, \qquad 0 \le P \le 1. \tag{4.10}$$

where I_{pol} is the intensity of the sum of the polarization components and I_{tot} is the total intensity of the beam. The value of $P = 1$ corresponds to completely polarized light, $P = 0$ corresponds to unpolarized light, and $0 < P < 1$ corresponds to partially polarized light.

Thus these parameters can be directly determined by the following six intensity measurements (I) performed with ideal polarizers: I_H, horizontal linear polarizer $(0°)$; I_V, vertical linear polarizer $(90°)$; I_P, $45°$ linear polarizer; I_M, $135°$ $(-45°)$ linear polarizer; I_R, right circular polarizer; and I_L, left circular polarizer.

$$\mathbf{S} = \begin{bmatrix} I \\ Q \\ U \\ V \end{bmatrix} = \begin{bmatrix} I_H + I_V \\ I_H - I_V \\ I_P - I_M \\ I_R - I_L \end{bmatrix}. \tag{4.11}$$

Table 4.1: Stokes vectors for different polarization states

State	H	V	P	M	R	L	Elliptical
$\mathbf{S}$	$\begin{pmatrix}1\\1\\0\\0\end{pmatrix}$	$\begin{pmatrix}1\\-1\\0\\0\end{pmatrix}$	$\begin{pmatrix}1\\0\\1\\0\end{pmatrix}$	$\begin{pmatrix}1\\0\\-1\\0\end{pmatrix}$	$\begin{pmatrix}1\\0\\0\\1\end{pmatrix}$	$\begin{pmatrix}1\\0\\0\\-1\end{pmatrix}$	$\begin{pmatrix}1\\\cos 2\alpha\\\sin 2\alpha\cos\delta\\\sin 2\alpha\sin\delta\end{pmatrix}$

The *Mueller matrix* provides a general framework for describing how an optical system modifies the polarization state of light. It relates the incident and transmitted Stokes vectors through a 4×4 real matrix and can represent fully polarized, partially polarized, and unpolarized light. Owing to its basis in measurable intensity quantities, the Mueller matrix formalism is widely used for polarization analysis and characterization of optical elements.

In matrix form this can be write as

$$\begin{pmatrix} S_0' \\ S_1' \\ S_2' \\ S_3' \end{pmatrix} = \begin{pmatrix} m_{00} & m_{01} & m_{02} & m_{03} \\ m_{10} & m_{11} & m_{12} & m_{13} \\ m_{20} & m_{21} & m_{22} & m_{23} \\ m_{30} & m_{31} & m_{32} & m_{33} \end{pmatrix} \begin{pmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{pmatrix} \tag{4.12}$$

or

$$\mathbf{S}' = \mathbf{M} \cdot \mathbf{S}. \tag{4.13}$$

4.3 | Medium Polarization Characteristics

A transparent optical medium is generally characterized by three fundamental polarization properties: diattenuation, depolarization, and retardance. In this experiment, the primary focus is on determining the retardance of the material. Retardance originates from the birefringent nature of a medium and produces a phase difference between orthogonal polarization components of light. This phase difference depends on the thickness and optical properties of the material. Retardance can be classified into two main types.

1. **Linear retardance:** Linear retardance arises from a phase delay between two *orthogonal linear polarization components*. It results from linear birefringence, which occurs due to anisotropy in the medium, leading to different refractive indices for orthogonal polarization directions. For a medium of thickness L, the linear retardance is given by

$$\delta = \frac{2\pi}{\lambda}(n_x - n_y)L, \tag{4.14}$$

   where λ denotes the wavelength of light, and n_x and n_y represent the refractive indices along two orthogonal axes.

2. **Circular retardance:** Circular retardance corresponds to *the phase difference between right- and left-circularly polarized light*. This effect originates from circular birefringence, typically present in optically active media lacking mirror symmetry. Such anisotropy produces different refractive indices for the two circular polarization states. For a propagation length L, the circular retardance is expressed as

$$\delta_c = \frac{2\pi}{\lambda}(n_R - n_L)L, \tag{4.15}$$

   where n_R and n_L are the refractive indices for right- and left-circularly polarized light, respectively. *Circular retardance manifests as optical rotation*, corresponding to the rotation of the plane of linear polarization about the propagation axis, with rotation angle

$$\psi = \frac{\delta_c}{2}. \tag{4.16}$$

4.4 | Spatial Light Modulator (SLM)

The *Spatial Light Modulator* (SLM) is an opto-electronic device used to control the spatial properties of a light beam by modulating its amplitude, phase, or polarization. Most commonly, SLMs are based on liquid crystal technology, where a pixelated layer of liquid crystals is placed between polarizing elements. When an external voltage is applied to each pixel, the orientation of the liquid crystal molecules changes, altering the polarization state of the transmitted light and thereby controlling its intensity or phase. Because each pixel can be addressed electronically through a computer, SLMs enable dynamic and real-time manipulation of optical wavefronts, allowing the generation of programmable amplitude and phase patterns for various applications in beam shaping, holography, and optical experiments.

Since the SLM is made of liquid crystals which rotate and align according to the voltage applied to them, the general form of the Mueller matrix of an SLM can be written as a retarder matrix multiplied with a rotation matrix. Now the Muller matrix for a retarder (or compensator) with fast axis toward x-axis can be written as

$$M_c = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos\delta & \sin\delta \\ 0 & 0 & -\sin\delta & \cos\delta \end{pmatrix} \tag{4.17}$$

For θ angle rotated wave plate we can write

$$M_c(\delta, 2\theta) = M_R(-2\theta) M_c M_R(2\theta) \tag{4.18a}$$

$$M_R(2\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 2\theta & \sin 2\theta & 0 \\ 0 & -\sin 2\theta & \cos 2\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{4.18b}$$

So we can write

$$M_c(\delta, 2\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos^2 2\theta + \cos\delta\ \sin^2 2\theta & (1-\cos\delta)\sin 2\theta\cos 2\theta & -\sin\delta\sin 2\theta \\ 0 & (1-\cos\delta)\sin 2\theta\cos 2\theta & \sin^2 2\theta + \cos\delta\ \cos^2 2\theta & \sin\delta\cos 2\theta \\ 0 & \sin\delta\sin 2\theta & -\sin\delta\cos 2\theta & \cos\delta \end{pmatrix}. \tag{4.19}$$

Now at the end for SLM we can write

$$M_{\text{SLM}} = M_c(\delta, 2\theta) \times M_R(-2\Psi) \tag{4.20a}$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos^2 2\theta + \sin^2 2\theta\cos\delta & \sin 2\theta\cos 2\theta(1-\cos\delta) & -\sin 2\theta\sin\delta \\ 0 & \sin 2\theta\cos 2\theta(1-\cos\delta) & \sin^2 2\theta + \cos^2 2\theta\cos\delta & \cos 2\theta\sin\delta \\ 0 & \sin 2\theta\sin\delta & -\cos 2\theta\sin\delta & \cos\delta \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 2\psi & -\sin 2\psi & 0 \\ 0 & \sin 2\psi & \cos 2\psi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{4.20b}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 2\theta\cos(2\theta-2\psi) + \sin 2\theta\cos\delta\sin(2\theta-2\psi) & -\cos\delta\sin 2\theta\cos(2\theta-2\psi) + \cos 2\theta\sin(2\theta-2\psi) & -\sin\delta\sin 2\theta \\ 0 & \sin 2\theta\cos(2\theta-2\psi) - \cos 2\theta\cos\delta\sin(2\theta-2\psi) & \cos\delta\cos 2\theta\cos(2\theta-2\psi) + \sin 2\theta\sin(2\theta-2\psi) & \cos 2\theta\sin\delta \\ 0 & \sin\delta\sin(2\theta-2\psi) & -\cos(2\theta-2\psi)\sin\delta & \cos\delta \end{pmatrix} \tag{4.20c}$$

Now as per the experiment once we take I_H polarized light and another time we take I_P polarized light. By applying the SLM matrix we can find

$$S_H = M_{\text{SLM}} \times \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ \cos 2\theta\cos(2\theta-2\psi) + \sin 2\theta\cos\delta\sin(2\theta-2\psi) \\ \sin 2\theta\cos(2\theta-2\psi) - \cos 2\theta\cos\delta\sin(2\theta-2\psi) \\ \sin(2\theta-2\psi)\sin\delta \end{pmatrix} \tag{4.21}$$

and,

$$S_P = M_{\text{SLM}} \times \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -\cos\delta\sin 2\theta\cos(2\theta - 2\psi) + \cos 2\theta\sin(2\theta - 2\psi) \\ \cos\delta\cos 2\theta\cos(2\theta - 2\psi) + \sin 2\theta\sin(2\theta - 2\psi) \\ -\cos(2\theta - 2\psi)\sin\delta \end{pmatrix} \tag{4.22}$$

So finally we can write

$$S_H(2) + S_P(3) = (1 + \cos\delta)\cos 2\psi = X_1, \tag{4.23a}$$
$$S_H(3) - S_P(2) = (1 + \cos\delta)\sin 2\psi = Y_1, \tag{4.23b}$$

$$\boxed{\delta = \cos^{-1}\left[\sqrt{X_1^2 + Y_1^2} - 1\right]} \tag{4.24}$$

$$\boxed{\psi = \frac{1}{2}\tan^{-1}\left(\frac{Y_1}{X_1}\right)}. \tag{4.25}$$

where δ is *linear retardance* and Ψ is *optical rotation*.

5 | Set up and Procedure

This experiment has two parts. In the first part, we did measure the angles of polarizers (P1 and P2) for Horizontal polarization. And in the later part we did Stokes polarimetry for SLM.

Part 1

1. First we had to find the *Brewster's angle* for the interface. In our case it was air-glass interface. As we know, at Brewster's angle the reflected beam becomes s-polarised (I_V), so we placed the P1 such that the output intensity is minimum. So we got the angle of P1 for I_H.

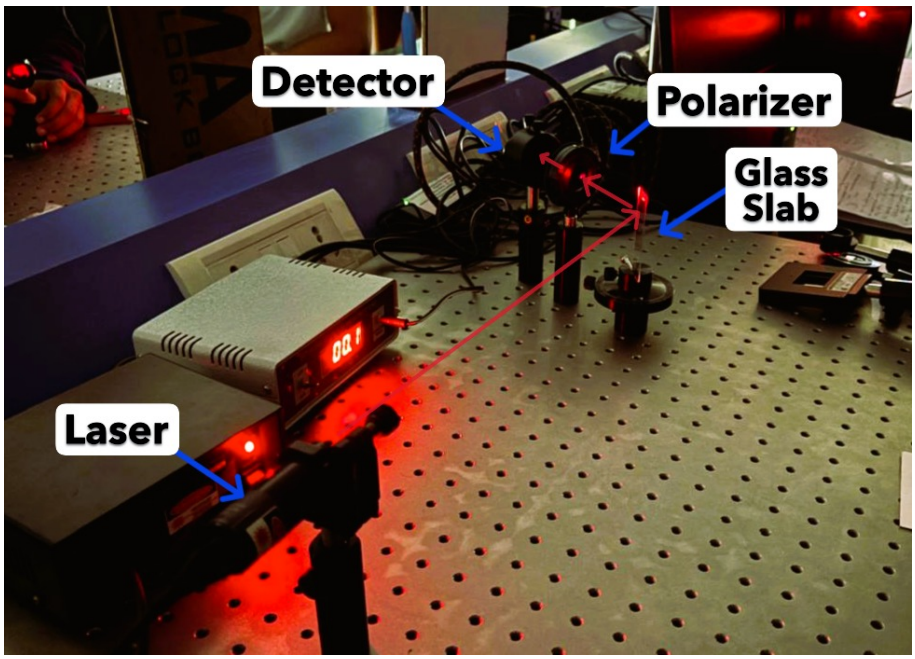


Figure 5.1: Setup to find the axis of the first polarizer (P1) using Brewster's angle approch

2. Similarly, for finding the polarization angle for P2, we placed this two polarizer in straight line and then kept the P1 at I_H and rotated the P2 such that we got minimum intensity. Thus we got the polarization angle for P2 also.

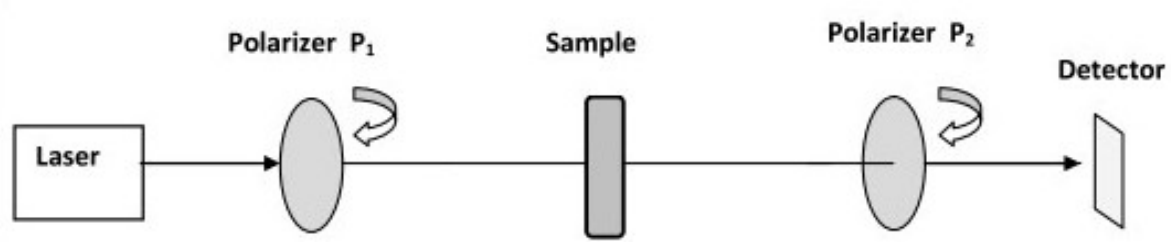


Figure 5.2: Schematic of the final setup

Part 2

In this part we configured the setup (see Figure 5.3) such that, the maximum measured intensity after the two polarizers does not exceed the μA scale of the photodetector.

1. The polarimetry was done for 4 different value of grey scale (applied voltage) value for SLM.
2. Keeping P1 at I_H and I_P, all stokes parameter were noted (after P2) for different grey scale (see Figure 5.3).

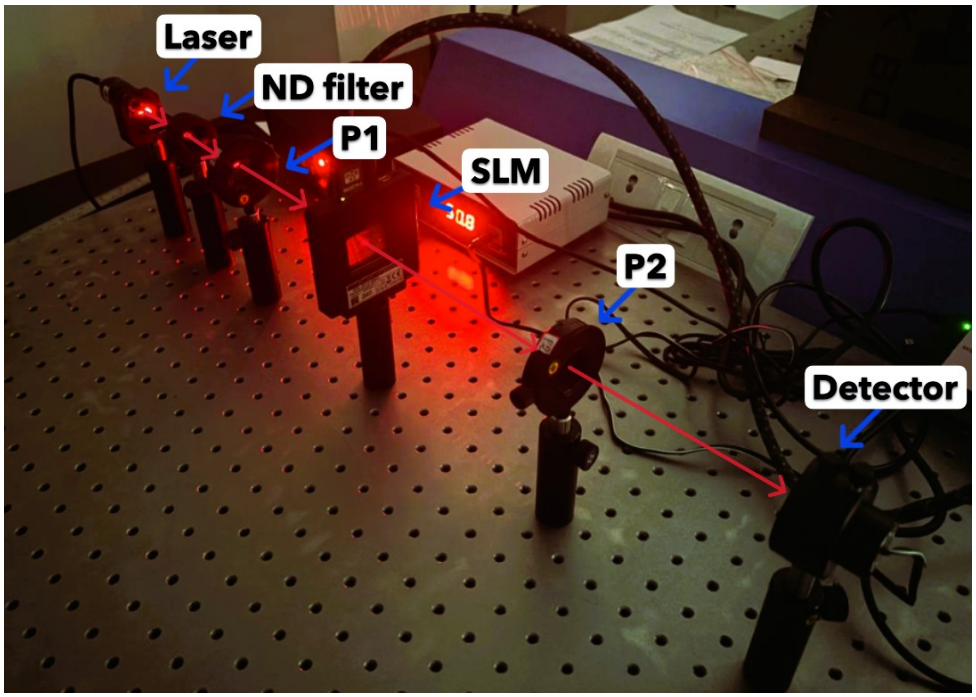


Figure 5.3: Final setup for Stokes polarimetry

6 | Data analysis and calculation

In the First part, we calculated the Brewster angle for the air-glass interface.

$$\theta_B = \tan^{-1} \frac{n_{glass}}{n_{air}} \tag{6.1a}$$

$$\theta_B = 56.3^\circ \tag{6.1b}$$

Now using the steps mentioned in *part 1*, we found the angles of the polarizer P1 and P2.

Polarizer	I_H	I_V	I_P	I_M
P1	308	38	353	263
P2	116	206	161	71

Table 6.1: Angles of polarization for P1 and P2

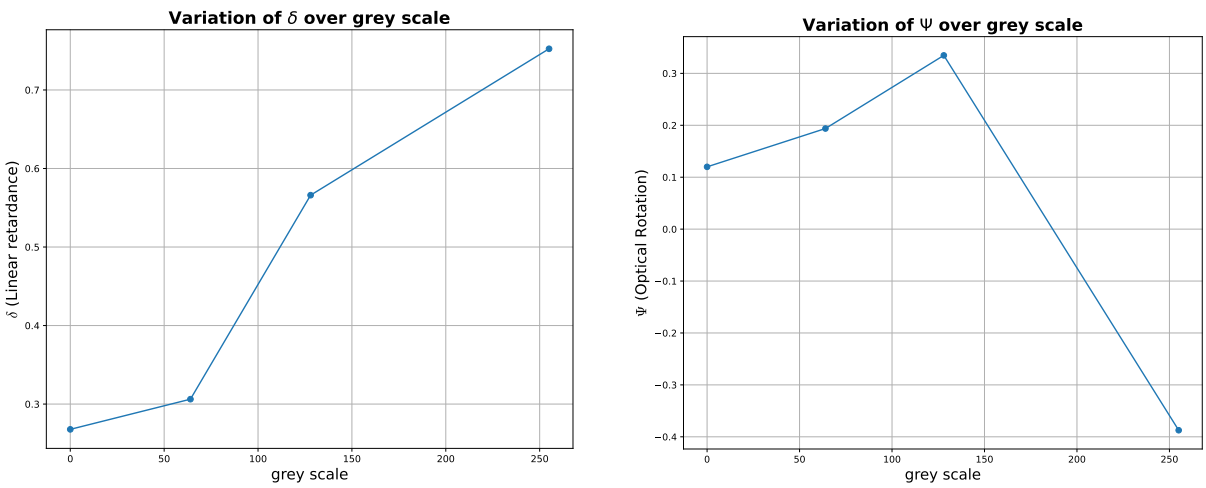
Now this angles were used for measurement in the second part. Using the mentioned method in part 2, the following measurements were done and tabulated.

Table 6.2: Measured photodetector currents for different grey levels and input polarization states.

Grey Level	Input polarization state	Intensities (Photodetector Current) (μA)			
		I_H	I_V	I_P	I_M
0	H	9.5	0.2	6.5	4.1
	P	56.2	87.8	139.8	3.1
64	H	10.1	0.4	8.1	3.8
	P	47.8	94.6	134	8
128	H	10.1	1.2	9.5	2.4
	P	36.1	113	124	25.8
255	H	2.5	8.4	8.8	2.1
	P	30.7	121.1	24	129.5

Using the above table we have calulcated the second and third components of stokes parameters S_H (Equation (4.21)) and S_P (Equation (4.22)).

Using the data and using Equation (4.23), Equation (4.24) and Equation (4.25), we plotted the variation of linear retardence and optical rotation over the grey scale value.



(a) Variation of δ (linear retardance)

(b) Variation of Ψ (optical rotation)

Figure 6.1: Variation over grey scale for SLM

7 | Error Calculation

All experimentally measured quantities possess uncertainties arising primarily from the finite resolution of the photodetector and alignment limitations of optical components. Since the detector resolution dominates over statistical fluctuations, the uncertainty analysis in this experiment was carried out using the **Maximum Permissible Error (MPE)** method.

The smallest measurable photodetector current in the experiment was

$$\Delta I = 0.1\ \mu\text{A}, \tag{7.1}$$

which is taken as the maximum uncertainty associated with all measured intensity values.

7.1 | Maximum Permissible Error Formula

For a function depending on measured variables,

$$f = f(x_1, x_2, \ldots, x_n), \tag{7.2}$$

the maximum propagated uncertainty is given by

$$\Delta f_{\max} = \sum_{i=1}^{n} \left| \frac{\partial f}{\partial x_i} \right| \Delta x_i. \tag{7.3}$$

This method provides an upper bound on the uncertainty by assuming that all measurement errors contribute in the same direction.

7.2 | Uncertainty in Normalised Stokes Parameters

The normalized Stokes parameters are

$$S(2) = \frac{I_H - I_V}{I_H + I_V}, \tag{7.4a}$$

$$S(3) = \frac{I_P - I_M}{I_H + I_V}. \tag{7.4b}$$

7.2.1 | Maximum Error in *S(2)*

Let

$$S(2) = \frac{I_H - I_V}{I_H + I_V}.$$

Using the MPE formula,

$$\Delta S(2)_{\max} = \left| \frac{\partial S(2)}{\partial I_H} \right| \Delta I + \left| \frac{\partial S(2)}{\partial I_V} \right| \Delta I. \tag{7.5}$$

The partial derivatives are

$$\frac{\partial S(2)}{\partial I_H} = \frac{2I_V}{(I_H + I_V)^2}, \tag{7.6a}$$

$$\frac{\partial S(2)}{\partial I_V} = \frac{2I_H}{(I_H + I_V)^2}. \tag{7.6b}$$

Hence,

$$\boxed{\Delta S(2)_{\max} = \frac{2\Delta I}{I_H + I_V}.} \tag{7.7}$$

7.2.2 | Maximum Error in *S(3)*

For

$$S(3) = \frac{I_P - I_M}{I_H + I_V},$$

$$\Delta S(3)_{\max} = \sum \left| \frac{\partial S(3)}{\partial I_i} \right| \Delta I. \tag{7.8}$$

Evaluating derivatives,

$$\frac{\partial S(3)}{\partial I_P} = \frac{1}{I_H + I_V}, \tag{7.9a}$$

$$\frac{\partial S(3)}{\partial I_M} = \frac{1}{I_H + I_V}, \tag{7.9b}$$

$$\frac{\partial S(3)}{\partial I_H} = \frac{|I_P - I_M|}{(I_H + I_V)^2}, \tag{7.9c}$$

$$\frac{\partial S(3)}{\partial I_V} = \frac{|I_P - I_M|}{(I_H + I_V)^2}. \tag{7.9d}$$

Thus,

$$\boxed{\Delta S(3)_{\max} = \frac{2\Delta I}{I_H + I_V} + \frac{2|I_P - I_M|}{(I_H + I_V)^2}\Delta I.} \tag{7.10}$$

7.3 | Uncertainty in *X* and *Y*

Applying MPE propagation,

$$\Delta X_{\max} = \Delta S_H(2)_{\max} + \Delta S_P(3)_{\max}, \tag{7.11}$$

$$\Delta Y_{\max} = \Delta S_H(3)_{\max} + \Delta S_P(2)_{\max}. \tag{7.12}$$

7.4 | Maximum Error in Optical Rotation

Optical rotation is given by

$$\psi = \frac{1}{2}\tan^{-1}\left(\frac{Y}{X}\right). \tag{7.13}$$

Using MPE propagation,

$$\Delta\psi_{\max} = \left|\frac{\partial\psi}{\partial X}\right| \Delta X + \left|\frac{\partial\psi}{\partial Y}\right| \Delta Y. \tag{7.14}$$

$$\frac{\partial\psi}{\partial X} = -\frac{Y}{2(X^2 + Y^2)}, \tag{7.15a}$$

$$\frac{\partial\psi}{\partial Y} = \frac{X}{2(X^2 + Y^2)}. \tag{7.15b}$$

$$\boxed{\Delta\psi_{\max} = \frac{|Y|\Delta X + |X|\Delta Y}{2(X^2 + Y^2)}.} \tag{7.16}$$

7.5 | Maximum Error in Linear Retardance

Linear retardance is calculated as

$$\delta = \cos^{-1}\left(\sqrt{X^2+Y^2}-1\right). \tag{7.17}$$

Let

$$Z = \sqrt{X^2+Y^2}-1.$$

Then,

$$\Delta\delta_{\max} = \left|\frac{d\delta}{dZ}\right|\Delta Z. \tag{7.18}$$

Since,

$$\frac{d}{dZ}\cos^{-1}(Z) = \frac{1}{\sqrt{1-Z^2}},$$

and

$$\Delta Z_{\max} = \frac{|X|\Delta X + |Y|\Delta Y}{\sqrt{X^2+Y^2}},$$

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$$\boxed{\Delta\delta_{\max} = \frac{|X|\Delta X + |Y|\Delta Y}{\sqrt{X^2+Y^2}\sqrt{1-(\sqrt{X^2+Y^2}-1)^2}}.} \tag{7.19}$$

7.6 | Numerical Estimation of Maximum Error

The detector resolution gives the uncertainty in all measured intensities as

$$\Delta I = 0.1\ \mu\text{A}. \tag{7.20}$$

A sample calculation of uncertainty propagation is shown for Grey Level $= 0$. Here are the measured intensities (normalised):

$$I_H = 9.5, \quad I_V = 0.2, \quad I_P = 6.5, \quad I_M = 4.1\ (\mu A)$$

Total intensity:

$$I = I_H + I_V = 9.7\ \mu A.$$

The normalized Stokes parameters are

$$S_H(2) = \frac{I_H - I_V}{I_H + I_V} = \frac{9.5-0.2}{9.7} = 0.959, \tag{7.21a}$$

$$S_H(3) = \frac{I_P - I_M}{I_H + I_V} = \frac{6.5-4.1}{9.7} = 0.247. \tag{7.21b}$$

Using the Maximum Permissible Error formula,

$$\Delta S_H(2)_{\max} = \frac{2\Delta I}{I_H + I_V}. \tag{7.22}$$

Substituting values,

$$\Delta S_H(2)_{\max} = \frac{2(0.1)}{9.7} = 0.0206. \tag{7.23}$$

The maximum uncertainty in $S_H(3)$ is

$$\Delta S_H(3)_{\max} = \frac{2\Delta I}{I_H + I_V} + \frac{2|I_P - I_M|}{(I_H + I_V)^2}\Delta I. \tag{7.24}$$

Substituting numerical values,

$$\begin{aligned}\Delta S_H(3)_{\max} &= \frac{0.2}{9.7} + \frac{2(2.4)}{(9.7)^2}(0.1) \\ &= 0.0206 + 0.0051 \\ &= 0.0257.\end{aligned} \tag{7.25}$$

For the $+45°$ input state,

$$I_H = 56.2, \quad I_V = 87.8, \quad I_P = 139.8, \quad I_M = 3.1.$$

Total intensity:

$$I = 144.0\ \mu A.$$

$$S_P(2) = \frac{56.2 - 87.8}{144.0} = -0.219, \tag{7.26a}$$

$$S_P(3) = \frac{139.8 - 3.1}{144.0} = 0.949. \tag{7.26b}$$

$$\begin{aligned}\Delta S_P(2)_{\max} &= \frac{2(0.1)}{144.0} = 0.00139, \\ \Delta S_P(3)_{\max} &= \frac{0.2}{144.0} + \frac{2(136.7)}{(144)^2}(0.1) \\ &= 0.00139 + 0.00132 \\ &= 0.00271.\end{aligned} \tag{7.27}$$

$$X = 0.959 + 0.949 = 1.908, \tag{7.28a}$$

$$Y = 0.247 - (-0.219) = 0.466. \tag{7.28b}$$

Maximum uncertainties:

$$\Delta X_{\max} = 0.0206 + 0.00271 = 0.0233, \tag{7.29a}$$

$$\Delta Y_{\max} = 0.0257 + 0.00139 = 0.0271. \tag{7.29b}$$

Error in Optical Rotation

$$\psi = \frac{1}{2}\tan^{-1}\left(\frac{Y}{X}\right).$$

$$\Delta\psi_{\max} = \frac{|Y|\Delta X + |X|\Delta Y}{2(X^2 + Y^2)}. \tag{7.30}$$

Substituting values,

$$\begin{aligned}\Delta\psi_{\max} &= \frac{(0.466)(0.0233) + (1.908)(0.0271)}{2(1.908^2 + 0.466^2)} \\ &= 0.0081 \text{ rad.}\end{aligned} \tag{7.31}$$

Error in Linear Retardance

$$\delta = \cos^{-1}\left(\sqrt{X^2 + Y^2} - 1\right).$$

$$\Delta\delta_{\max} = \frac{|X|\Delta X + |Y|\Delta Y}{\sqrt{X^2 + Y^2}\sqrt{1 - (\sqrt{X^2 + Y^2} - 1)^2}}. \tag{7.32}$$

Substituting numerical values,

$$\Delta\delta_{\max} = 0.056 \text{ rad.} \tag{7.33}$$

The same procedure was followed for all other grey levels to obtain the maximum uncertainties in linear retardance and optical rotation.

8 | Sources of Error

The uncertainties in the determination of linear retardance and optical rotation arise from several experimental and instrumental limitations. The major sources of error in the present experiment are listed below.

1. **Photodetector Resolution:** The dominant source of uncertainty originates from the finite resolution of the photodetector. The smallest measurable current was $0.1\ \mu$A, which limits the precision of all intensity measurements and directly propagates into the calculated Stokes parameters.
2. **Laser Intensity Fluctuations:** Temporal fluctuations in the laser output power lead to variations in measured intensities during successive measurements, affecting the accuracy of normalized Stokes parameters.
3. **Polarizer Misalignment:** Small angular misalignment of the input polarizer (P1) and analyzer (P2) from their intended orientations introduces systematic errors in the measured polarization components. The least count of this angle is $2°$.
4. **Finite Extinction Ratio of Polarizers:** Real polarizers do not completely block the orthogonal polarization component. Leakage of unwanted polarization reduces measurement contrast and modifies the evaluated Stokes parameters.
5. **Mechanical Rotation Stage Error:** Limited angular resolution and backlash in the rotation mounts can produce slight deviations from the desired polarization angles.
6. **SLM Pixel Non-uniformity:** The liquid crystal pixels of the SLM may not respond uniformly to the applied grey level, causing spatial variation in phase retardation across the beam.
7. **Beam Alignment and Positioning Errors:** Imperfect alignment of the laser beam through the optical components may lead to partial clipping or non-uniform illumination of the SLM surface. During our experiment one of the screw which hold the laser was missing.
8. **Background and Ambient Light:** Stray ambient light and detector dark current contribute additional signal offsets, especially for low-intensity measurements.

9 | Discussion

The variation of linear retardance (δ) and optical rotation (ψ) with grey level of the spatial light modulator (SLM) was determined using Stokes vector measurements. The obtained plots show clear dependence of the polarization properties of the SLM on the applied grey scale value, which controls the voltage across the liquid crystal layer. While the overall behaviour follows the expected physical mechanism of liquid crystal modulation, noticeable deviations from an ideal theoretical trend are observed. The experimental results are discussed below.

- **Expected behaviour of linear retardance:** In a nematic liquid crystal SLM, the applied grey level determines the effective voltage across the liquid crystal molecules. Increasing voltage causes the molecules to progressively align along the electric field direction, thereby reducing the effective birefringence experienced by the incident light. *The retardance is theoretically expected to decrease approximately monotonically with increasing grey level. The experimental plot shows an overall variation consistent with voltage-controlled phase modulation, although the change is not perfectly monotonic.* Such deviations arise because the electro–optic response of liquid crystals is inherently nonlinear, and the grey level does not correspond linearly to the applied voltage due to internal device calibration and digital driving electronics.
- **Expected behaviour of optical rotation:** Optical rotation originates from circular birefringence and depends on the orientation of the liquid crystal director relative to the incident polarization state. Unlike linear retardance, optical rotation does not necessarily vary monotonically with applied voltage. As the liquid crystal molecules reorient, the coupling between orthogonal polarization components changes in a complex manner, which can produce increases, decreases, or near-zero values of rotation at different grey levels. *Therefore, the*

non-monotonic variation observed in the optical rotation plot is consistent with the expected behaviour of liquid crystal polarization modulation rather than an indication of experimental failure.

- **Nonlinear voltage–phase relationship:** The grey scale values applied to the SLM are digitally mapped to driving voltages through internal electronics. This mapping is typically nonlinear (gamma correction), resulting in a nonlinear relationship between grey level and phase retardation. Consequently, equal steps in grey level do not produce equal changes in retardance, leading to departure from an ideal smooth curve.
- **Imperfect polarization preparation and alignment:** The theoretical analysis assumes perfectly defined input polarization states and ideal analyzer orientations. In practice, finite extinction ratios of polarizers and slight angular misalignment introduce residual polarization components, which modify the measured Stokes parameters and affect the extracted values of retardance and optical rotation.
- **SLM pixel non-uniformity:** The liquid crystal alignment across the SLM surface is not perfectly uniform. Since the detector measures the averaged transmitted intensity over many pixels, spatial variations in phase modulation lead to systematic deviations from the behaviour expected for an ideal uniform retarder.

Despite quantitative deviations from the ideal theoretical behaviour, the experimental results clearly demonstrate that both linear retardance and optical rotation depend on the applied grey level, confirming voltage-controlled manipulation of polarization by the liquid crystal SLM. The observed trends are therefore consistent with the fundamental operating principle of a phase-modulating spatial light modulator.

10 | Conclusion

In this experiment, the polarization characteristics of a spatial light modulator (SLM) were studied using Stokes vector polarimetry. The linear retardance and optical rotation were determined from measured intensity data for different grey levels applied to the SLM. The results show that the retardance varies systematically with grey level while the optical rotation displays a non-monotonic behaviour consistent with changes in molecular orientation.

Deviations from ideal theoretical behaviour are attributed to the nonlinear response of the liquid crystal, alignment imperfections, and finite detector resolution. Overall, the experiment successfully demonstrates voltage-controlled polarization modulation by the SLM and confirms the usefulness of the Stokes–Mueller formalism for characterizing polarization-modifying optical elements.

11 | References

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